

Title: Bus Route and Schedule Tracker using GPS

Description: This project develops a system to monitor buses in real time using GPS, manage routes and schedules, and inform passengers about current location, next stop, and estimated arrival time. It can be designed for a campus shuttle, school bus, or city transport service.

Problem Statement: Passengers often do not know where the bus is or when it will arrive. A GPS-based tracker solves this by combining live location updates with stored route and schedule data.

Scenario: A student selects the campus shuttle route and stop. The system checks the bus GPS position, determines the next stop, calculates the remaining distance, and displays an estimated arrival time such as “Bus T818 will arrive at the library in 4 minutes”.

Team and Scope of the Project: Suitable for a group of up to 5 students. Work can be divided into route design, schedule handling, GPS simulation, search/ETA calculation, and user interface/testing.

Data structures can be used: Graph – bus stops and routes, Queue – upcoming stops, Linked List – ordered route path, Hash Table – fast bus/stop lookup, and Priority Queue – nearest-arrival buses.